

Code Challenges

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#=====08-04-2026=====#  
1 : Even or Odd Function  
Create a function check_number()  
Ask the user to enter a number  
Use try-except to handle invalid input  
Use if statement to check:  
even  
odd  
Print the result  
def check_number():  
    user = int(input("enter a number:"))  
    try :  
        if user%2==0:  
            print("even number")  
        else:  
            print("odd numbers")  
    except Exception as e:  
        print("invalid error")  
check_number()  
2 : Sum of Numbers Using Loop  
Ask the user how many numbers they want to enter  
Use a loop to take numbers  
Handle invalid input using try-except  
Find the total sum  
Print the result  
def sum_1():  
    total=0  
    try:  
        user = int(input("enter how many numbers:"))  
        for i in range(user):  
            n=int(input("enter a numbers:"))  
            total=total+n  
            print("total",total)  
        except Exception as e:  
            print("invalid input")  
sum_1()  
3 : Multiplication Table  
Create a function table()  
Ask the user to enter a number  
Handle invalid input  
Use a for loop to print multiplication table from 1 to 10  
def function_1():  
    try:  
        user = int(input("enter a number:"))  
        for i in range(1,11):  
            # print(user,"x",i,"=",user*i)
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        print(f"{user} x{i}={user*i}")
    except Exception as e:
        print("invalid input",e)
function_1()
4 : Positive Number Checker
Create a function positive()
Ask the user for a number
Use if statement:
if negative → raise exception
Handle the error using except
Print suitable message
def function_1():
    try:
        user = int(input("enter a number:"))
        if user<0:
            raise ValueError("negative numbers not allowed")
        print("positive number")
    except Exception as e:
        print("invalid input",e)
function_1()
5 : Module Function
Create a module mymodule.py
Add a function add(a, b)
Import the module in another file
Ask the user for two numbers
Handle invalid input using try-except
Call the module function and print result
import mymodule
try:
    a = int(input("enter a numbers:"))
    b = int(input("enter a numbers:"))
    print(mymodule.add_1(a,b))
except Exception as e:
    print("invalid valid",e)
6 : Login System
Create a correct password variable
Use a loop to allow 3 attempts
Use if statement to check password
Print:
"Login successful"
or "Wrong password"
Use try-except if input causes error
def corr_1():
    password="Learn"
    count =3
    try:
        while count>0:
            user = input("enter the password:")

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        if user==password:
            print("login successful")
            break
        else:
            count-=1
            print("wrong password")
    except Exception as e:
        print("invalid input",e)
corr_1()
7 : List Average Program
Create a function average()
Use loop to take 5 numbers from user
Store them in a list
Handle invalid input
Find average and print it
def function_ave():
    n=[]
    while len(n)<5:
        try:
            user=int(input("enter a number:"))
            n.append(user)
        except Exception as e:
            print("invalid input",e)
    total=sum(n)
    avg=total/len(n)
    print("average=",avg)
function_ave()
8 : Student Marks Module
Create a module student.py
Add function to calculate total marks
Import module into another file
Ask user for marks using loop
Handle invalid input
Print total and average
import mymodule
marks = []
try:
    for i in range(3):
        user= int(input("enter a marks:"))
        marks.append(user)
    total = mymodule.student_1(marks)
    average = total/len(marks)
    print("total:",total)
    print("Average:",average)
except Exception as e:
    print("invalid input",e)
9.Odd and Even Counter
Create a function count_numbers()

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Ask the user to enter 5 numbers using loop
Use if statement to count:
even numbers
odd numbers
Handle invalid input using try-except
Print total even and odd numbers
def count_number():
    even=0
    odd=0
    try:
        for i in range(5):
            user=int(input("enter a number:"))
            if user%2==0:
                even+=1
            else:
                odd+=1
        print("even numbers:",even)
        print("odd numbers:",odd)
    except Exception as e:
        print("invalid input",e)
count_number()

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10.Largest Number Finder

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Create a function largest()
Ask user to enter 3 numbers
Use if-elif to find largest number
Handle invalid input
Print the largest number
def largest_1():
    try:
        big=0
        for i in range(3):
            num=int(input("enter a number:"))
            if i==0:
                big=num
            elif num>big:
                big=num
        print("largest number :",big)
    except Exception as e:
        print("invalid input",e)
largest_1()

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11.Countdown Program

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Create a function countdown()
Ask user for starting number
Use while loop to count down to 1
Handle invalid input
Print "Time Over" at end
def countdown():
    try:

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        user = int(input("enter a number:"))
        while user>=1:
            print(user)
            user-=1
        print("Time over")
    except Exception as e:
        print("invalid input",e)
countdown()

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12.Simple Attendance System

Create a **list** of student names

Use loop to display names

Ask user to search a student

Use **if** statement to check availability

Handle errors using **try-except**

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def system_1():
    list_1=["priya","raju","rahul","rani"]
    try:
        for i in list_1:
            user = input("enter a name:")
            if user == i:
                print("its available")
                break
            else:
                print("not available")
    except Exception as e:
        print("invalid input",e)
system_1()

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13.Marks and Result

Create a function result()

Ask user **for** marks of 5 subjects using loop

Find total **and** average

Use **if-elif** for:

Pass

Fail

Grade

Handle invalid input

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def result_1():
    total=0
    try:
        for i in range(5):
            marks=int(input("enter a number:"))
            total+=marks
        average=total/5
        print("total:",total)
        print("average:",average)
        if average>=35:
            print("pass")
        elif average>=90:

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        print("grade:A")
    else:
        print("fail")
except Exception as e:
    print("invalid input",e)
result_1()
14.Number Guessing Game
Import random module
Generate random number from 1 to 20
Use loop to guess number
Print:
Too high
Too low
Correct
Handle invalid input
import random
num = random.randint(1,20)
while True:
    try:
        user = int(input("guess the number:"))
        if user>num:
            print("too high")
        elif user<num:
            print("too low")
        else:
            print("correct")
            break
    except Exception as e:
        print("invalid input",e)
15.Random Password Generator
Import random module
Create a function password()
Generate random 4-digit password
Use loop to allow 3 attempts
Use if statement for correct/wrong password
Handle invalid input
import random
def password():
    pword = "Test"
    count = 3
    try:
        while True:
            user = input("enter a password:")
            if user==pword:
                print("correct password")
                break
            elif count==0:
                print("your chances over")

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        else:
            print("wrong password")
            count-=1
    except Exception as e:
        print("invalid input",e)
password()
16.Shopping Cart
Create a function cart()
Use loop to add item prices
Stop loop when user enters "stop"
Find total bill
Handle invalid inputs
def cart():
    total=0
    while True:
        try:
            price=input("enter the price of the item:")
            total+=price
            if price=="stop":
                break
            total+=price
        except Exception as e:
            print("invalid input",e)
    print("total bill:",total)
cart()

#=====Mean, Median, Mode=====#
num = [10,20,30,40]
mean = sum(num)/len(num)
print("mean:",mean)
1 : Mean Program
Create a list of numbers
Find total using sum()
Find mean using formula
Print mean
x=[1,2,3,4,5,6,7,8,9,10]
total = sum(x)
mean=total/len(x)
print("mean:",mean)
2 : User Input Mean
Ask user to enter 5 numbers using loop
Store them in list
Find mean
Handle invalid input
def user_1():
    list_1=[]
    try:
        for i in range(5):

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        n = int(input("enter a number:"))
        list_1.append(n)
        mean = sum(list_1)/len(list_1)
        print("mean:",mean)
    except Exception as e:
        print("invalid input",e)

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user_1()

3 : Median Finder

Create a **list** of numbers

Sort the **list**

Find middle value

Print median

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def median_1():
    x=[10,30,40,50,20]
    x.sort()
    medain = x[len(x)//2]
    print("medain is:",medain)

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median_1()

4 : Mode Finder

Create a **list** with repeated numbers

Find most repeated number

Print mode

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def mode_finder():
    x=[10,20,30,10,20,30,40]
    mode = max(x,key=x.count)
    print("mode is :",mode)

```

mode_finder()

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def mode_finder():
    x=[10,20,30,10,20,30,40]
    mode=0
    count=0
    for i in x:
        if x.count(i)>count:
            count=x.count(i)
            mode=i
    print("mode:",mode)

```

mode_finder()

5 : Mean, Median, Mode Program

Ask user to enter numbers

Store them **in list**

Find:

Mean

Median

Mode

Handle invalid input

Print all results

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def finding_1():
    x=[]

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try:
    for i in range(5):
        user = int(input("enter a numbers:"))
        x.append(user)
    mean = sum(x)/len(x)
    x.sort()
    median =x[len(x)//2]
    mode= max(x,key=x.count)
    print("mean:",mean)
    print("median:",median)
    print("mode:",mode)
except Exception as e:
    print("invalid input",e)
finding_1()
Final Challenge:
Create a function statistics()
Ask user for 10 numbers using loop
Calculate:
Mean
Median
Mode
Use:
loop
if statement
try-except
Print all outputs properly
def statistics_1():
    x=[]
    try:
        for i in range(10):
            user=int(input("enter a numbers:"))
            x.append(user)
        mean = sum(x)/len(x)
        x.sort()
        median = (x[4]+x[5])/2
        count=0
        mode=0
        for i in x:
            if x.count(i)>count:
                count = x.count(i)
                mode=i
        print("mean:",mean)
        print("medain:",median)
        print("mode:",mode)
    except Exception as e:
        print("invalid input",e)
statistics_1()

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